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AI for Bharat Hackathon

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Team Name : Spectacular Crew

Team Leader Name : Sriram Karanam

Problem Statement : AI for Retail, Commerce & Marketplace Ecosystems

Brief about the Idea:

Retail and marketplace businesses operate in highly dynamic environments where **seasonal demand shifts, inventory imbalance, pricing errors, and seller underperformance** can quickly lead to revenue loss and poor customer experience. Most existing retail analytics systems are reactive — they report what has already happened — or focus narrowly on demand forecasting without accounting for downstream business risks.

We propose an **AI-powered Retail Failure Simulator & Market Intelligence Platform** that enables **pre-emptive decision-making** by simulating retail failure scenarios such as seasonal demand mismatch, inventory overstock or stockout, and promotion inefficiencies. The system analyzes how these failures propagate across inventory, pricing, fulfilment, and revenue, and delivers **clear, actionable insights** to retail leaders through a **mobile-first executive experience**.
This solution helps retailers move from reactive firefighting to **proactive, risk-aware market intelligence and decision support**.

How is it different from existing ideas?

- Existing retail solutions focus on **demand forecasting or dashboards**
- Our solution focuses on **failure-first market intelligence**
- Treats seasonal demand changes as **business risk scenarios**, not just trends
- Simulates **cascading impact** across inventory, pricing, fulfilment, and revenue
- Provides **executive-ready insights via mobile**, not analyst-heavy dashboards
- AI is used for **reasoning, explanation, and trade-off analysis**, not blind automation

How will it solve the problem?

- Detect seasonal and market-driven demand shifts early
- Simulate failure scenarios such as overstock or stockout
- Quantify business impact (revenue loss, spoilage, missed sales)
- Understand *why* the risk is occurring through AI explanations
- Take proactive mitigation actions such as procurement adjustment, targeted promotions, or inventory reallocation

USP of the Proposed Solution

- Failure-First Market Intelligence (pre-mortem analysis for retail)
- Seasonal Demand Failure Simulation
- Cascading Impact Analysis (demand → inventory → revenue)
- AI-Driven Executive Explanations
- Mobile-First Decision Flow for Retail Leaders
- Human-in-the-Loop, Responsible AI Design

List of Features Offered by the Solution

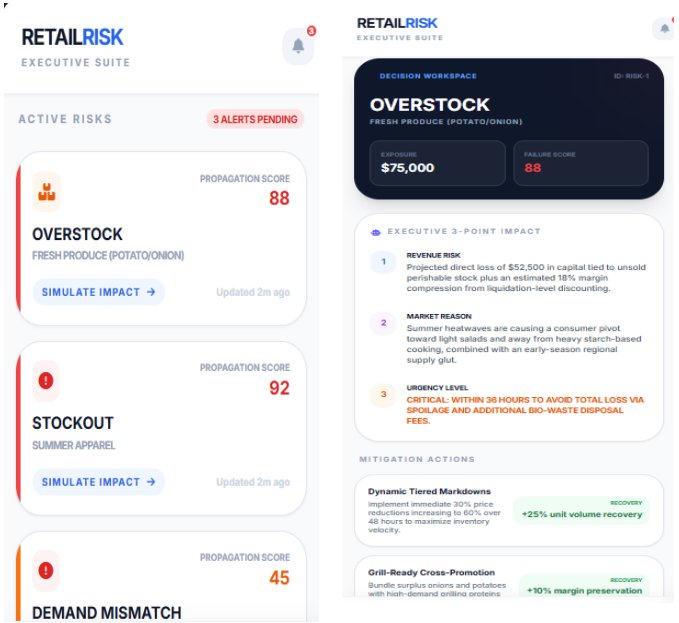
- Seasonal demand risk detection
- Inventory overstock & stockout simulation
- Retail Failure Propagation Score (risk spread metric)
- AI-generated 3-point executive impact summaries
- Market-aware mitigation recommendations
- Web dashboard for retail planners and analysts
- Mobile app for executives with alert-driven decision flow
- Explainable, auditable decision support

Mobile App → Risk Alert Notification → Three-Point Impact Summary → View Mitigation Options

Revenue risk
Market reason
Urgency level

Mitigation options:

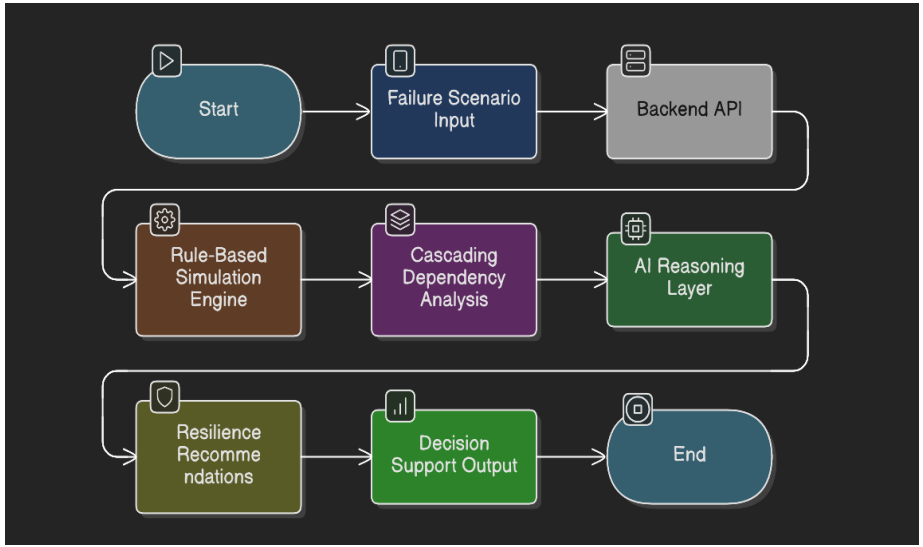
Reduce procurement
Promote high-demand seasonal items
Reallocate inventory regionally
This enables **decision-making in under 30 seconds**, without complex dashboards.



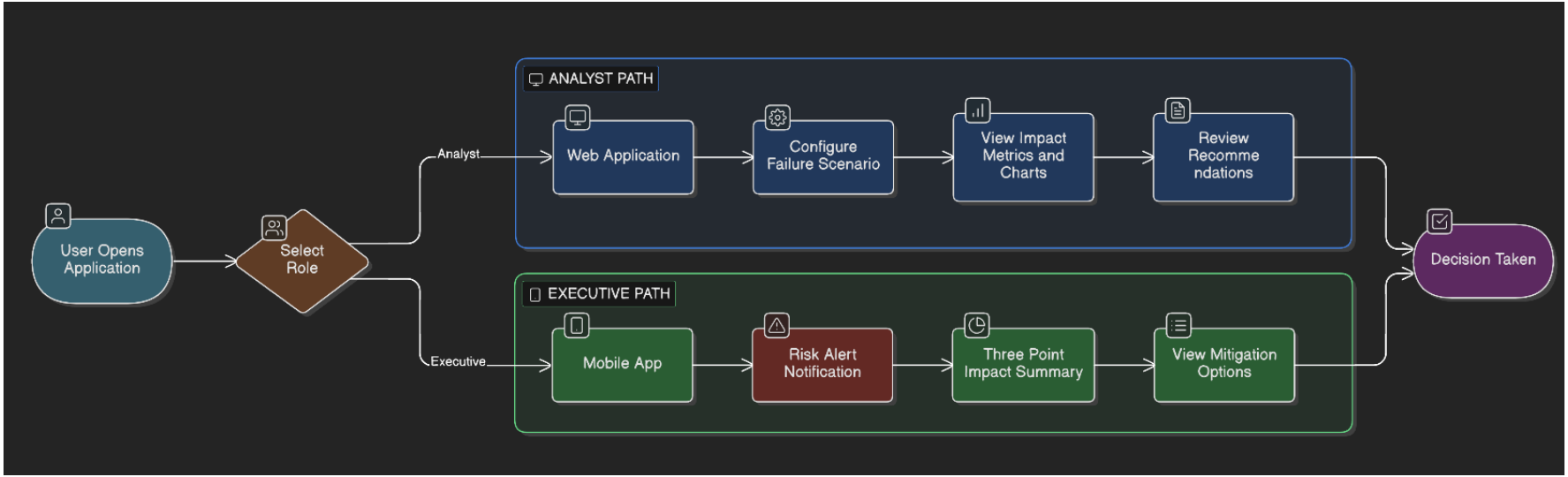
Sample prototype

Process flow diagram or Use-case diagram

1. Market data and seasonal trends are analyzed
2. Potential retail failure scenarios are detected
3. Failure simulation evaluates cascading business impact
4. AI reasoning layer explains causes and risk drivers
5. Mitigation strategies are generated with trade-offs
6. Insights are delivered via web dashboard and mobile app

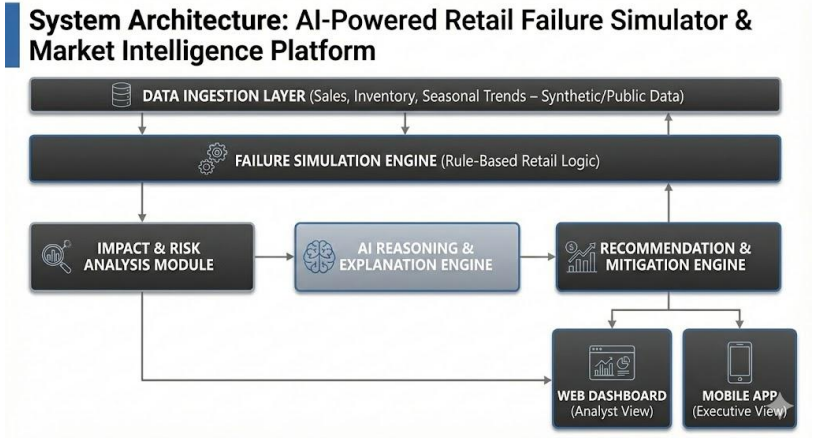


Wireframes/Mock diagrams of the proposed solution



Architecture diagram of the proposed solution:

- Data Ingestion Layer (sales, inventory, seasonal trends – synthetic/public data)
- Failure Simulation Engine (rule-based retail logic)
- Impact & Risk Analysis Module
- AI Reasoning & Explanation Engine
- Recommendation & Mitigation Engine
- Web Dashboard (Analyst View)
- Mobile App (Executive View)



Technologies to be used in the solution:

- AWS (Cloud infrastructure)
- Python (FastAPI / backend services)
- AI/ML (LLM-based reasoning)
- Data Processing (Pandas, NumPy)
- Frontend (Streamlit / React for web, Flutter mobile-responsive UI)
- Visualization (Charts & summary views)

Estimated implementation cost

- Low-cost prototype using synthetic data and managed AI services
- Incremental scaling based on retail adoption and integrations
- Designed for efficient experimentation before production rollout

Requirements for the hackathon:

- Uses **synthetic and public datasets only**
- AI supports decisions; humans remain in control
- Focused prototype to demonstrate depth, clarity, and feasibility
- Strong alignment with retail, commerce, and market intelligence goals

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Thank You